Name: Dilip S

Registration number: 25BAI10590

Degree: BTECH

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INVESTMENT MANAGEMENT SYSTEM

Financial literacy is increasingly important for young professionals, Having a tool to monitor investments helps in making informed decisions but Manual tracking in spreadsheets is error-prone and time-consuming a database-backed solution ensures data persistence and scalability. The problem addressed was the need for a personal investment tracker that could manage multiple asset classes (stocks, mutual funds, and bonds) in one place. Many students and early investors struggle to keep track of their investments, calculate totals, and update records efficiently.

Project Objective:

The primary goal of this project is to create a Python code to track and manage diverse financial assets Such as Stocks, Mutual Funds, and Bonds. The system aims to automate the calculation of investment values and maintain a persistent record of transactions using a MySQL database.

System Architecture:

Backend: A MySQL database named INVESTMENTS containing three primary tables:

- STOCKS
- MUTUAL_FUNDS
- BONDS

Frontend: A command-line interface in python that uses a nested menu system for user navigation.

Data Processing: Pandas library is used to structure SQL data into DataFrames for analysis.

Data Presentation: At last tabulate library is used to store data in a clean, readable grid format.

Database Structure:

The system automatically initializes the following table structures:

Table Name	Attributes	Data Types
STOCKS	Name, Quantity, BuyPrice	VARCHAR, INT, FLOAT
MUTUAL_FUNDS	FundName, Amount, MaturityDate	VARCHAR, FLOAT, FLOAT
BONDS	BondName, Quantity, FaceValue, CouponRate, MaturityDate	VARCHAR, INT, FLOAT, FLOAT, DATE

Queries for creating the tables:

- CREATE TABLE IF NOT EXISTS STOCKS (Name VARCHAR(30), Quantity INT, BuyPrice FLOAT)

- CREATE TABLE IF NOT EXISTS MUTUAL_FUNDS (FundName VARCHAR(50), amount FLOAT, MaturityDate FLOAT)

- CREATE TABLE IF NOT EXISTS BONDS (BondName VARCHAR(50), Quantity INT, FaceValue FLOAT, CouponRate FLOAT, MaturityDate DATE)

Implementation Details:

Stock Management:

The stock module includes an awesome Update feature. When a user adds more units of an existing stock, the system retrieves the previous data and calculates a new **weighted average cost.**

Mutual Funds and Bonds:

Mutual Funds: Focuses on lumpsum investments, tracking the principal amount and the investment horizon in years.

Bonds: Tracks fixed-income securities, recording specific coupon rates and maturity dates.

Total Valuation:

The total() function iterates through all three asset classes to calculate the total value of investments.

Error Handling:

Module Verification:

The program checks for required libraries (mysql.connector, pandas, tabulate) and provides installation instructions if they are missing.

Database Safety:

Uses parameterized queries (e.g., %s placeholders) to prevent SQL injection and ensure data integrity during insertion and deletion.

Input Validation:

Wrapped in try-except blocks to handle unexpected user inputs or database connection errors gracefully.

Challenges Faced:

- Handling invalid user inputs.
- Ensuring data consistency when updating stock quantities and prices.
- Designing a menu system that was intuitive yet flexible.
- Debugging SQL queries and managing commits to prevent data loss.

What I Learned:

- Practical use of **Python with MySQL** for CRUD operations.
- Importance of **exception handling** to make programs user-friendly.
- How to calculate **weighted averages** for stock prices.

- Value of **clean output formatting** (tabulate, pandas) in improving user experience.

- The iterative nature of development: refining code based on feedback and testing.

Future Enhancements:

Real-Time Market Integration:

Currently, The system relies on manual BuyPrice inputs. Integration of a financial API (like Yahoo Finance or Alpha Vantage) to:

- Live Tracking:
  Fetch current market prices to calculate Current Value vs Invested Value.

- Profit/Loss Analysis:
  Automatically calculate unrealized gains or losses for each stock and bond in your portfolio.

Advanced Portfolio Features:

- SIP (Systematic Investment Plan) Tracker:
  Enhance the MUTUAL_FUNDS logic to handle recurring monthly investments rather than just lumpsums.

- Dividend Tracker:
  Add a feature to log and calculate dividends earned from stocks and coupon payments from bonds.

- Currency Support:
  Add support for multiple currencies with real-time conversion rates for international investments.

Conclusion:

This project successfully integrates Python's analytical power with MySQL's storage capabilities. It provides a functional tool for personal finance management while maintaining clean code through modular functions and comprehensive error handling.

This project provided hands-on experience in combining database management with Python programming to solve a real-world problem. It strengthened skills in structured coding, error handling, and user interface design. The final product is a functional investment tracker that can be extended further with features like live market price integration, graphical dashboards, or export options.

Github link : https://github.com/dilip10590/Fundamentals-of-AIML-project.git

Reference:

- Vityarthi – vityarthi.com
- Github – github.com
- Googlecolab – googlecolab.com
- Peer feedbacks from:

  Gokul prasad k - 25BAI10148

  Bavesh p – 25BAI10093

  Sudharshan S G – 25BAI10706

  Raghul S K – 25BAI10288

THANK YOU